Kriging in Sports

CMDA-4654 Project 2

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Data Introduction

Points per shot is one of the biggest current topics of discussion in basketball analytics. It is calculated using two factors: a team or players chance of making a shot from a position on the floor and the point of value of a make from that position. Performing analysis on this topic requires gathering both of those values which can be done using the `hoopr` package.

```r
library(hoopR)

pbp <- load_nba_pbp(seasons = 2026)
player_box <- load_nba_player_box(seasons = 2026)

player_names <- player_box |>
  distinct(athlete_id, athlete_display_name)

shots <- pbp |>
  filter(shooting_play == TRUE,
         !is.na(coordinate_x), !is.na(coordinate_y), !grepl("Free Throw", type_text),
         season_type == 2) |>
  select(athlete_id_1, coordinate_x, coordinate_y, score_value) |>
  left_join(player_names, by = c("athlete_id_1" = "athlete_id")) |>
  mutate(
    x = ifelse(coordinate_x > 0,
               -(coordinate_x - 46.75),
               coordinate_x + 46.75),
    y = ifelse(coordinate_x > 0, -coordinate_y, coordinate_y)) |>
  filter(x <= 45) |>
  select(athlete_display_name, score_value, x, y)
```

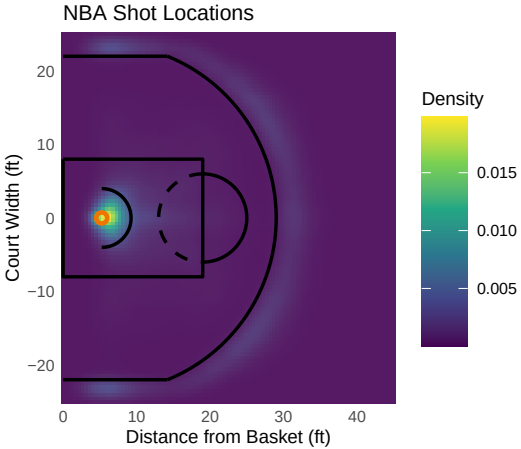
Data Example

The data has been loaded and saved as csv for ease of access: Click here to access shots.csv.

Preview of the data

athlete_display_name	score_value	x	y
Luka Garza	3	28	12
Franz Wagner	3	30	13
Luka Garza	0	24	17
Paolo Banchero	2	7	1
Baylor Scheierman	3	38	-1
Jalen Suggs	0	17	23
Ron Harper Jr.	0	27	15
Franz Wagner	3	4	-23
Baylor Scheierman	0	31	-8
Wendell Carter Jr.	0	15	23
Baylor Scheierman	0	8	23
Paolo Banchero	0	6	-3
Paolo Banchero	0	8	2
Paolo Banchero	0	5	0
Max Shulga	0	15	-2
Jordan Walsh	0	23	-21
Baylor Scheierman	0	10	-2
Ron Harper Jr.	2	8	-4
Ron Harper Jr.	2	7	-4
Desmond Bane	0	34	12

NBA Regular Season Shot Locations (2025-26):



Kriging Motivation

It is clear from the shot density layout on the previous slide that there are two main areas that teams take shots from:

the three point line
the rim

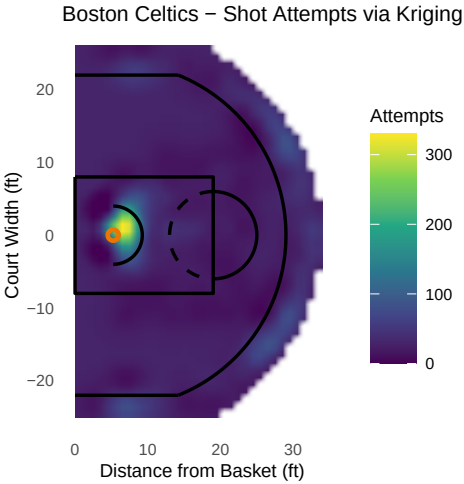
This makes sense given that shots at the rim are likely to be high percentage and shots from three point range are worth more points. The middle area between these two hot zones is fairly empty, and the league has been trending to make midrange jumpshots even less a part of offenses in the Modern Era. All of this evolution is in the pursuit of offensive effiency which is where Kriging comes in.

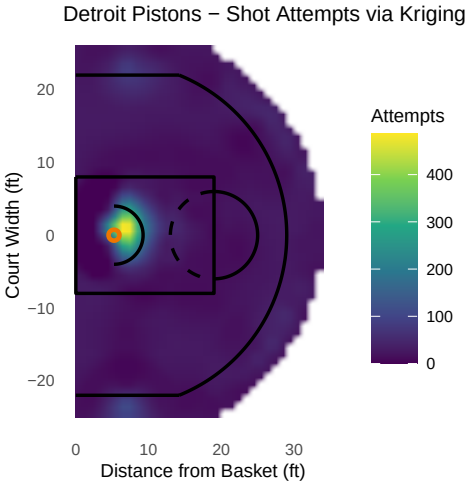
Since the shot map is made up of x and y data, it can be treated similarly to a geostatistics field from more typical Kriging - it has high density areas and low gaps with a large number of very individualized spots. Kriging can be used to smooth over these less commonly attempted areas and give a picture of not only where teams shoot but how well they perform from these areas.

Combining those two areas will mean that Kriging can make a points per shot estimation from every location on the floor which can then be used to check if teams are running their offenses to peak effiency.

Attempt Frequency Kriging

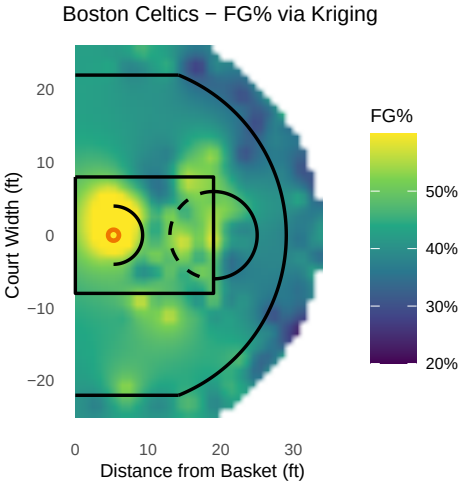
A typical picture of attempt frequency for a basketball team would consist of colored boxes or hexes that represent volume from each area of the court, but a custom Kriging can smooth this out into a continuous map using two foot by two foot bins.

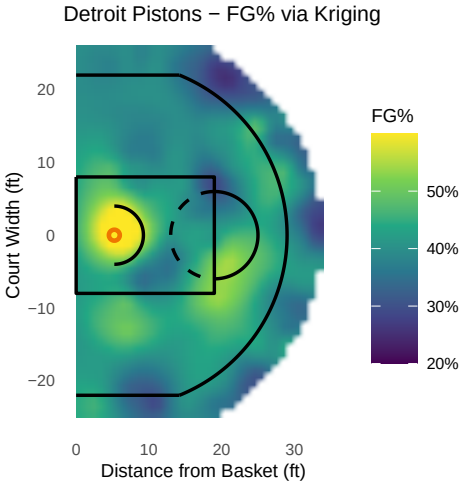




FG% Kriging

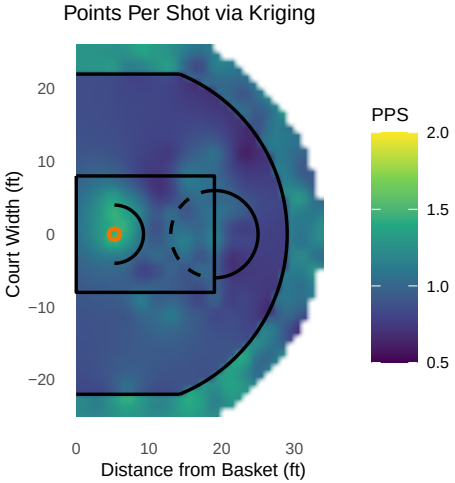
While the Kriging for shot attempts is somewhat trivial, it is extremely useful for field goal percentage. This statistic leads to a map that has continuous values across all the bins and the maps will differ more greatly team-to-team.

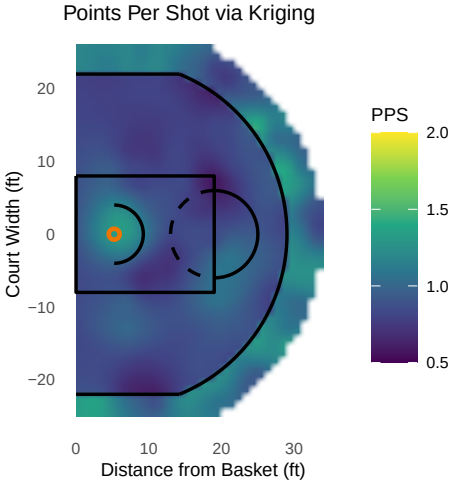




Points Per Shot

As mentioned previously, points per shot is a metric that is used to determine the quality of a type of shot attempt. Now that the field goal percentage Kriging has supplied a likelihood of making a shot at any location on the floor, it is easy to take it a step further into point values of the shots.



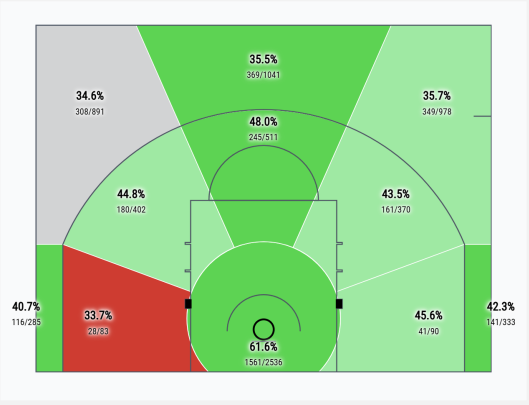


Non-Kriging Approach

So the question becomes "How do these smooth maps help NBA teams improve their offenses?"

One of the main advantages that these plots provide is the way they make it much easier to full picture of slots across the floor. As a comparison, here would be a more standard look at a teams percentages and shot volumes from Synergy Sports.

The plot makes it hard tell whether there's smaller deviations within the large groups, doesn't calculate the point values of the shots (i.e. shooting 50% from 2 is the same as 33% from 3).



Kriging Comparison

Accuracy of Predictions

Conclusions & Kriging Limitations

References